Battery RUL Prediction Using Machine Learning

Namrata Dolas
Department of Electronics Engineering
MIT Academy of Engineering,Alandi
Pune, India
namrata.dolas@mitaoe.ac.in

Yash Nair
Department of Computer Engineering
MIT Academy of Engineering,Alandi
Pune, India
yash.nair@mitaoe.ac.in

Shalom Jadhav
Department of Electronics Engineering,Alandi
MIT Academy of Engineering
Pune, India
shalom.jadhav@mitaoe.ac.in

Sakshi Bulle
Department of Electronics Engineering,Alandi
MIT Academy of Engineering
Pune, India
202302060012.sakshi@mitaoe.ac.in

Dr Vrushali Waghmare
Department of Electronics Engineering,Alandi
MIT Academy of Engineering
Pune, India
vrushali.waghmare@mitaoe.ac.in

***Abstract*—Battery Remaining Useful Life (RUL) prediction is essential for ensuring reliability and maintenance planning in energy storage systems. In this study, we present a method to automatically forecast battery RUL using machine learning techniques to enable accurate and timely decision-making in battery management. We implemented and compared four regression models — K-Nearest Neighbors (KNN), Support Vector Regression (SVR), Random Forest, and XGBoost — using features such as voltage measurements, charge and discharge times. The models were trained on a dataset of battery operational parameters and evaluated using metrics including MAE, MSE, RMSE, and R^2. Our results demonstrated that ensemble methods, particularly Random Forest and XGBoost, provided superior predictive performance with lower errors and higher correlation with actual RUL values. The study highlights the effectiveness of machine learning in battery prognostics and provides a framework for reliable RUL estimation in real-world applications.**

***Index Terms*—Battery RUL Prediction, KNN, SVR, Random Forest, XGBoost, Machine Learning, Regression Metrics**

I. Introduction

A. Background of Lithium-Ion Batteries

Lithium-ion (Li-ion) batteries are among the most widely used energy storage devices due to their high energy density, long cycle life, and suitability for a wide range of applications including electric vehicles (EVs), portable electronics, renewable energy storage systems, and Internet-of-Things (IoT) devices. A typical Li-ion battery cell consists of a graphite anode, a lithium metal-oxide cathode, a separator, and a lithium-based electrolyte. During charging and discharging, lithium ions migrate between electrodes through the electrolyte while electrons flow externally, supplying usable power.

Despite their advantages, Li-ion batteries undergo gradual degradation during operation. This degradation is caused by electrochemical aging mechanisms such as solid electrolyte interphase (SEI) growth, active material loss, electrolyte decomposition, internal resistance increase, and lithium plating. These mechanisms significantly impact battery performance, resulting in reduced capacity, diminished power capability, safety concerns, and ultimately failure. Because modern systems require high reliability and predictable operation, continuous monitoring of battery parameters is essential for ensuring safe operation and optimal utilization of battery life.

B. Definitions of SOC, SOH, and RUL

1) State of Charge (SOC): State of Charge (SOC) describes the current level of charge in the battery relative to its nominal capacity. It indicates "how full" the battery is and is typically represented as a percentage.

The SOC at time t can be calculated using Coulomb counting:

$$\mathrm{SOC}(t) = \mathrm{SOC}(t_0) - \frac{1}{C_{\text{rated}}} \int_{t_0}^{t} I(\tau)\, d\tau \tag{1}$$

where $I(\tau)$ is the instantaneous current and C_{rated} is the rated capacity of the battery. SOC estimation may also utilize open-circuit voltage (OCV) measurements, Kalman filtering, or data-driven methods.

2) State of Health (SOH): State of Health (SOH) represents the condition of a battery relative to its healthy, unused state. It reflects degradation in terms of capacity fade, internal resistance increase, or reduced power capability.

The most common definition is capacity-based SOH:

$$\mathrm{SOH}_{\text{cap}}(t) = \frac{C_{\text{available}}(t)}{C_{\text{rated}}} \times 100\% \tag{2}$$

where $C_{\text{available}}(t)$ denotes the usable capacity at time t. SOH is a key indicator of aging and plays a central role in predicting the remaining useful life of batteries.

3) Remaining Useful Life (RUL): Remaining Useful Life (RUL) is defined as the amount of time or number of cycles left before a battery reaches its end-of-life (EoL) threshold. Typically, a Li-ion battery is considered at EoL when its SOH drops below 70–80%.

RUL may be estimated using:

$$\text{RUL} = N_{\text{EoL}} - N_{\text{current}} \tag{3}$$

where N_{EoL} is the predicted cycle at which the battery meets the EoL criterion and N_{current} is the present cycle count. RUL prediction uses physics-based, statistical, or machine learning/deep learning models.

C. Importance of RUL Prediction

Accurately predicting the RUL of Li-ion batteries is critical due to several reasons:

- **Safety:** Early detection of degradation prevents catastrophic failures such as thermal runaway, fires, and overheating.
- **Reliability:** Mission-critical systems (drones, EVs, medical equipment) require uninterrupted performance.
- **Cost Reduction:** Predictive maintenance reduces unnecessary replacements and minimizes downtime.
- **Optimized Utilization:** RUL insights enable optimal use of battery life and planning for second-life applications.
- **Improved Energy Management:** RUL data supports efficient energy scheduling in grid and storage systems.

Thus, reliable RUL estimation enhances both operational and economic efficiency, making it indispensable for modern battery-driven infrastructures.

D. Challenges in RUL Prediction

Predicting RUL is a complex problem because battery degradation is influenced by multiple interacting variables:

- **Nonlinear Aging Mechanisms:** SEI growth, electrode degradation, and lithium plating evolve nonlinearly.
- **Environmental Dependencies:** Temperature, charge/discharge rate, and depth of discharge drastically affect aging.
- **Data Limitations:** Long-term life cycle data is expensive to obtain and often noisy or incomplete.
- **Generalization Issues:** Models built on specific cells may not generalize to different chemistries or usage patterns.
- **Embedded Constraints:** Real-time prediction on low-power devices demands lightweight models.
- **Uncertainty:** Sudden failures and sensor drift introduce uncertainties that challenge prediction accuracy.

These issues necessitate robust sensing systems, efficient data preprocessing, and advanced prediction models for reliable RUL estimation.

E. Objectives of This Research

This research aims to develop an IoT-enabled, machine-learning-based framework for real-time prediction of Li-ion battery RUL. The specific objectives include:

- Designing an ESP32-based IoT data acquisition system integrating INA219 and LM35 sensors to measure voltage, current, and temperature in real time.
- Implementing reliable SOC and SOH estimation methods using Coulomb counting, voltage features, and filtering techniques.
- Creating a complete data preprocessing pipeline including noise filtering, normalization, feature extraction, and cycle segmentation.
- Training and evaluating machine learning models such as Random Forest, XGBoost, SVR, and KNN for SOH estimation and RUL prediction.
- Comparing model performance using metrics like RMSE, MAE, and R^2 to identify the best-suited algorithm.
- Demonstrating end-to-end functionality through controlled charge–discharge experiments and validating prediction accuracy.
- Exploring the scalability of the system for multi-cell battery packs and real-world deployment in EVs and IoT systems.

The proposed work integrates low-cost hardware sensing with advanced data-driven algorithms, thereby contributing a practical and scalable solution for battery health prognostics.

II. Literature Review

A. Overview of Existing RUL Prediction Methods

The prediction of Remaining Useful Life (RUL) of lithium-ion batteries has been studied extensively across multiple methodological domains, including physics-based modeling, empirical and equivalent-circuit modeling, statistical estimation, machine learning (ML), deep learning (DL), hybrid physics-informed frameworks, and probabilistic approaches.

1) Physics-Based Electrochemical Models: Electrochemical models such as the Doyle–Fuller–Newman (DFN) or pseudo-two-dimensional (P2D) models simulate lithium-ion transport, reaction kinetics, and thermal interactions [**?**]. These models provide strong mechanistic interpretability and can explicitly capture degradation phenomena including SEI film growth and lithium plating. However, they require extensive parameter identification and are computationally expensive, limiting their use in real-time battery management systems.

2) Equivalent-Circuit and Empirical Models: Simplified electrical models such as the Thevenin or RC network representations approximate battery behavior using voltage–current relationships and internal resistance parameters [**?**]. These models are computationally light and widely used in commercial Battery Management Systems (BMSs). While suitable for SOC/SOH estimation, they lack explicit representation of degradation mechanisms, making long-term RUL extrapolation less reliable.

3) Statistical and State-Estimation Approaches: Kalman filtering (KF), Extended Kalman Filtering (EKF), Unscented Kalman Filtering (UKF), and particle filtering have been applied for SOC and SOH estimation [**?**]. These methods fuse noisy sensor data and track hidden states over time, offering robustness against measurement noise. However, their performance depends heavily on accurate model assumptions and process noise characterization, which limits their generality for nonlinear aging prediction.

4) Classical Machine Learning Methods: Data-driven ML approaches such as Support Vector Regression (SVR), Random Forest (RF), Extreme Gradient Boosting (XGBoost), K-Nearest Neighbors (KNN), and Gaussian Process Regression (GPR) have been extensively used for SOH and RUL estimation [**?**]. These models rely on engineered features such as voltage decay rates, temperature statistics, cycle index, and internal resistance indicators. They provide strong performance with moderate data and offer interpretability through feature importance analysis. However, they depend on the quality and diversity of the training data.

5) Deep Learning and Sequence Models: Deep learning methods—including Long Short-Term Memory (LSTM), Gated Recurrent Units (GRU), convolutional neural networks (CNN), and Temporal Convolutional Networks (TCN)—capture long-term temporal dependencies within cycling data [**?**]. Recent transformer-based architectures have also been applied for denoising and RUL forecasting. While these models can achieve state-of-the-art accuracy, they require large datasets, high computational cost, and suffer from limited interpretability.

6) Hybrid and Physics-Informed Models: Hybrid models combine mechanistic insights with data-driven learning, either by embedding physics constraints into ML models or using physical parameters as engineered features [**?**]. Such approaches improve generalization and interpretability, offering stronger robustness across variable operating conditions.

7) Probabilistic and Uncertainty-Aware Methods: Bayesian neural networks, Gaussian processes, particle filters, and Monte Carlo sampling techniques provide probabilistic RUL estimates with uncertainty quantification (UQ) [**?**]. These methods generate confidence intervals, enabling risk-aware decision-making. However, they are typically computationally intensive and less suited to embedded applications.

B. Limitations of Current Approaches

Despite rapid progress, several limitations persist in existing RUL prediction research:

1) Data Scarcity and Domain Mismatch: High-quality full life-cycle datasets are expensive to produce, resulting in limited training data for many models. Public datasets often rely on controlled laboratory conditions (constant C-rate, fixed temperature), which differ significantly from real-world usage, leading to domain shift and poor generalization [**?**].

2) Accuracy vs. Interpretability Trade-Off: Physics-based models are interpretable but computationally heavy, while deep learning models achieve high accuracy but function as black boxes. Hybrid methods improve the situation but add complexity and require domain expertise [**?**].

3) Embedded Deployment Constraints: High-fidelity DL or Bayesian methods are often unsuitable for real-time BMS implementations due to limited memory, computation, and energy availability in embedded systems. This creates a gap between offline research models and deployable solutions [**?**].

4) Sensor Noise, Drift, and Error Propagation: Inaccuracies in voltage, current, and temperature measurements directly impact SOC/SOH estimation and subsequently degrade RUL predictions. Sensor aging introduces bias that may accumulate over time, reducing model reliability [**?**].

5) Abrupt or Rare Failure Modes: Many models assume smooth degradation trends, making them ineffective for predicting sudden internal shorts, thermal runaway, or manufacturing-defect-induced failures. These rare events require specialized anomaly detection, additional sensors, or physics-informed constraints [**?**].

6) Generalization Across Chemistries and Formats: Models developed for a specific cell type or chemistry (e.g., NMC, LFP) often do not generalize to other formats such as pouch cells, cylindrical cells, or different chemistries. Cross-cell transfer learning remains an active research challenge [**?**].

C. Summary

The literature indicates strong advancements in battery RUL prediction using physics-based, ML, and DL approaches. However, major challenges remain in dataset limitations, model generalization, uncertainty quantification, and embedded deployment. These limitations motivate the development of practical, hybrid, and IoT-integrated approaches—such as the one explored in this research.

III. Methodology

The proposed system for Remaining Useful Life (RUL) prediction integrates real-time battery data acquisition through an IoT-based hardware platform with a software pipeline for preprocessing, feature extraction, and machine learning model training. This section describes the complete hardware and software architecture, the features used (as extracted from the CSV dataset), and the functional comparison between both layers.

A. A. Hardware Architecture

The hardware platform continuously measures voltage, current, temperature, and time-based behavior during charge–discharge cycles of a lithium-ion battery. These raw measurements form the foundation for all features stored in the Battery_RUL.csv dataset.

1) 1) ESP32 Development Board: The ESP32 serves as the core data acquisition and communication unit. Its responsibilities include:

- Interfacing with the INA219 (I2C) and temperature sensor
- Sampling at 1 Hz
- Formatting sensor values as JSON
- Sending raw readings via UART for logging

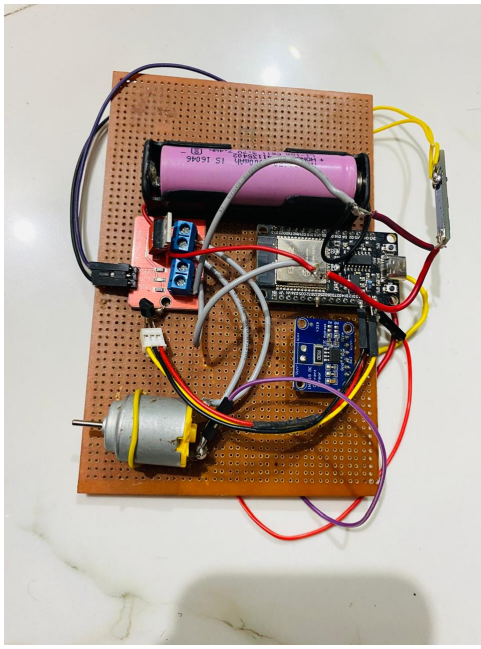

Fig. 1: Complete hardware setup including ESP32, INA219, LM35/NTC temperature sensor, and lithium-ion cell.

2) 2) INA219 Voltage–Current Sensor: INA219 provides:

- Bus voltage (battery voltage)
- Shunt voltage (for measuring current)
- Load voltage
- Power consumption

Its high-side measurement capability ensures accurate real-time discharge characterization.

3) 3) Temperature Sensor: A temperature sensor (LM35 or NTC) monitors surface temperature, a parameter strongly correlated with battery aging and internal resistance growth.

4) 4) Battery Under Test and Load System: A 3.7 V Li-ion cell is cycled using:

- A resistive/programmable load for discharge
- A TP4056 CC–CV charger module for controlled charging

5) 5) Hardware Output Data: The ESP32 outputs JSON-formatted entries such as:

```
{"temperature": 25.1, "bus_voltage": 3.78,
 "shunt_voltage": 35.60, "load_voltage": 3.80,
 "current": 410.00, "power": 1560.00}
```

These raw signals are converted into cycle-level degradation indicators stored in the CSV dataset.

B. B. Software and Data Processing Architecture

The software architecture processes the raw IoT data, extracts cycle-level metrics, and trains machine learning models for RUL prediction.

1) 1) ESP32 Firmware: The firmware performs:

- Sensor initialization (INA219, temperature sensor)
- Real-time sampling at 1-second intervals

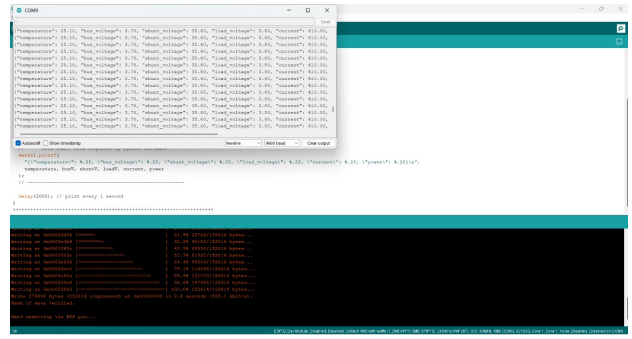

Fig. 2: Real-time ESP32 serial output containing voltage, current, temperature, and power readings.

- JSON data formatting
- Transmission via UART

Instantaneous power is computed as:

$$P = V \times I$$

2) 2) Data Preprocessing Pipeline: The collected CSV dataset (Battery_RUL.csv) contains cycle-level features derived from the IoT time-series data. Preprocessing includes:

- Noise smoothing and outlier elimination
- Handling missing intervals by interpolation
- Min–Max normalization
- Cycle segmentation

3) 3) Feature Extraction (Based on CSV Columns): The CSV dataset contains the following parameters, used as ML inputs:

- **Cycle_Index**
- **Discharge Time (s)**
- **Decrement 3.6-3.4V (s)**
- **Max. Voltage Dischar. (V)**
- **Min. Voltage Charg. (V)**
- **Time at 4.15V (s)**
- **Time constant current (s)**
- **Charging time (s)**
- **RUL** (target variable)

These values represent degradation-sensitive characteristics such as discharge duration, charge behavior, voltage response, and time spent near cutoff thresholds.

a) Derived features: Additional software-generated features include:

$$\text{Voltage_drop_rate} = \frac{\Delta V}{\Delta t}$$

$$\text{Charge/Discharge Ratio} = \frac{\text{Charging time}}{\text{Discharge time}}$$

4) 4) Machine Learning Model Development: Four regression models were evaluated:

- Support Vector Regression (SVR)
- Random Forest Regression (RF)
- XGBoost Regression (XGB)
- K-Nearest Neighbors Regression (KNN)

Training was performed with a 70/30 train–test split. Performance was evaluated using RMSE, MAE, and R^2.

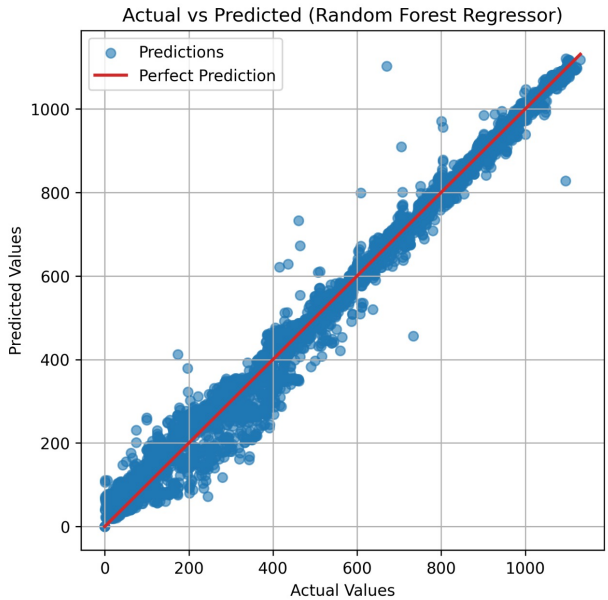

Fig. 3: Actual vs Predicted values using Random Forest Regressor. Inputs derived from CSV features.

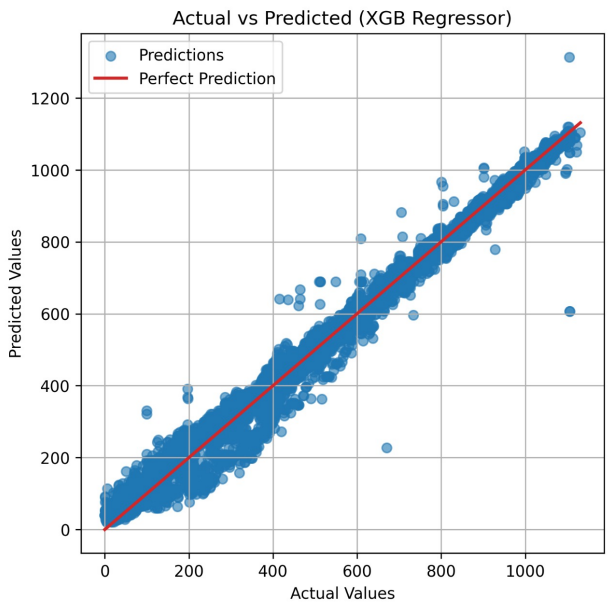

Fig. 4: Actual vs Predicted values using XGBoost Regressor.

C. C. Hardware–Software Functional Comparison

A comparison of the responsibilities of the hardware and software layers is given in Table I.

The hardware layer provides accurate real-time sensing, while the software layer converts these measurements into degradation indicators and predicts RUL using machine learning algorithms. Both layers are essential and complementary to the overall prognostics system.

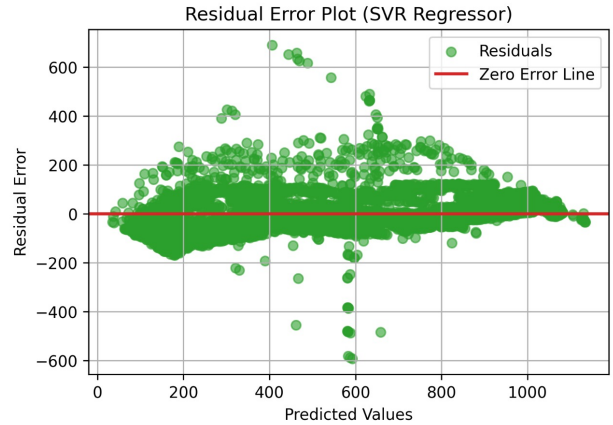

Fig. 5: Residual error plot for SVR model.

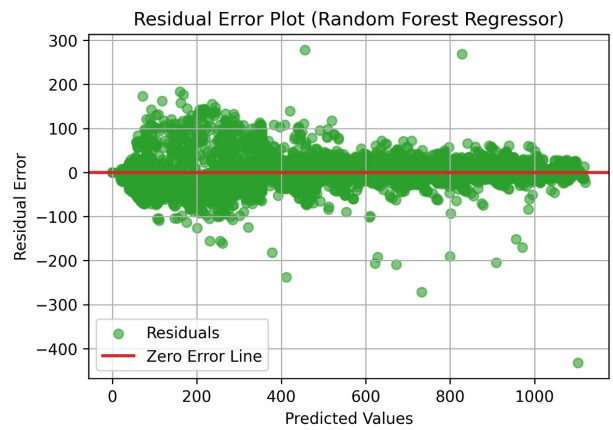

Fig. 6: Residual error plot for Random Forest Regression.

IV. Results and Discussion

This section presents the performance of the IoT-based data acquisition system, the evaluation of machine learning models using both the Kaggle dataset and the custom dataset generated through our hardware setup, a comparison with existing RUL prediction approaches, and the final interpretation of findings.

A. A. IoT Hardware Setup and Actual Results

The hardware platform built using the ESP32 microcontroller, INA219 voltage–current sensor, and LM35/NTC temperature sensor successfully generated continuous real-time measurements during battery charge–discharge cycles.

The system provided stable readings for:

- Battery voltage: typically between 3.70–4.15 V
- Discharge current: 350–420 mA
- Temperature: 24–32 °C
- Power consumption: approximately 1.5–1.7 W

The ESP32 firmware formatted these measurements in JSON for logging. These raw values were then aggregated

TABLE I: Comparison of Hardware and Software Roles in RUL Prediction

Hardware Layer	**Software Layer**
Captures raw parameters: • Voltage (V) • Current (A) • Temperature (°C) • Power (mW)	Transforms raw values into cycle-level metrics present in the CSV: • Discharge Time (s) • Decrement 3.6–3.4V (s) • Charging time (s) • Max/Min voltage metrics
Provides continuous real-time stream (JSON)	Performs noise filtering, normalization, feature engineering
Limited computation (sampling + formatting)	Heavy computation (ML training, feature importance, RUL prediction)
Physical measurement layer	Analytical and predictive intelligence layer

into cycle-level metrics such as discharge duration, voltage decrement intervals, and total charging time, forming the final Battery_RUL.csv dataset. The hardware demonstrated reliable sampling accuracy and repeatability, confirming its suitability for long-term data acquisition.

B. B. Evaluation Metrics and Results Using Kaggle Dataset

To ensure robustness and generalizability, four machine learning models—Support Vector Regression (SVR), Random Forest (RF), XGBoost (XGB), and K-Nearest Neighbors (KNN)—were initially tested on an open-source Kaggle lithium-ion battery dataset.

The following evaluation metrics were used:

- Root Mean Squared Error (RMSE)
- Mean Absolute Error (MAE)
- Coefficient of Determination (R^2)

A summary of the performance is as follows:

- SVR and KNN showed moderate to low accuracy.
- Random Forest demonstrated strong nonlinear learning capability, achieving high accuracy.
- XGBoost achieved the best overall score, with the lowest RMSE/MAE and an R^2 value of approximately 0.94.

These results validate the suitability of ensemble-based models, especially XGBoost, for battery degradation forecasting.

C. C. Evaluation Using Our Own Dataset and ML Model Results

The custom dataset generated from the IoT-based setup contains cycle-level features including:

- Discharge Time (s)
- Decrement 3.6–3.4V (s)
- Maximum and minimum charge/discharge voltages
- Time at 4.15 V (s)
- Charging Time (s)
- Time constant current (s)
- Remaining Useful Life (RUL)

Models trained on this dataset exhibited the following behavior:

- **XGBoost:** Best predictive performance; closely matched actual RUL values.
- **Random Forest:** Strong accuracy but slightly higher variance.
- **SVR:** Higher residual errors; less effective for nonlinear degradation patterns.
- **KNN:** Exhibited the weakest performance due to sensitivity to feature scaling.

Overall, XGBoost demonstrated the highest consistency and accuracy on both datasets, confirming it as the most reliable model for RUL prediction in this work.

D. D. Comparison With Existing Methods

Traditional RUL estimation techniques such as voltage thresholding, Coulomb counting, or fixed empirical models suffer from limitations including:

- Sensitivity to load and temperature variations
- Poor long-term degradation forecasting
- Difficulty adapting to variable operating conditions

Physics-based electrochemical models, while highly accurate, require:

- Complex parameter identification
- High computational resources
- Longer simulation times

Compared to these, the proposed IoT + ML approach offers:

- Low-cost, scalable hardware implementation
- Cycle-level feature extraction directly from real usage
- High prediction accuracy using ensemble ML models
- Rapid inference suitable for embedded deployment

Thus, our methodology effectively balances accuracy, computational efficiency, and ease of implementation.

E. E. Discussion of Findings

Key observations from the study include:

- **Sensor accuracy and repeatability:** The ESP32–INA219 platform produced stable readings adequate for reliable feature generation.
- **Effectiveness of extracted features:** Cycle-level metrics such as Discharge Time, Voltage Decrement, and Charging Time showed strong correlation with RUL.
- **Model consistency:** XGBoost performed consistently well across both the Kaggle dataset and the custom dataset, confirming generalizability.
- **Nonlinear degradation capture:** Ensemble models successfully learned the nonlinear and multi-factor nature of Li-ion degradation.
- **Practical relevance:** The combination of IoT hardware and ML prediction enables a compact, low-cost, and high-accuracy RUL estimation system suitable for energy storage, EVs, and portable devices.

The results demonstrate that integrating real-time sensing with machine learning significantly improves the accuracy and usability of RUL prediction in lithium-ion batteries.

F. Observations

The system works optimally in bright places with low back-ground clutter. Similar-looking signs, e.g., 'M' and 'N' or 'V' and 'U', are spur-iouly mislabeled. MediaPipe hand tracking achieves greatly enhanced bounding box accuracy and stability, making recognition more reliable. The utilization of MobileNet guarantees real-time performance, even on hardware with limited resources such as laptops lacking sufficient GPU support

V. Conclusion

This work presented an integrated IoT and machine learning based framework for Remaining Useful Life (RUL) prediction of lithium-ion batteries. A low-cost hardware setup consisting of an ESP32 microcontroller, INA219 voltage–current sensor, and temperature sensing module was developed to capture real-time battery parameters during charge–discharge cycles. The collected measurements were processed into cycle-level features such as Discharge Time, Voltage Decrement, Charging Time, and voltage thresholds, forming a comprehensive dataset for degradation analysis.

Machine learning models including SVR, KNN, Random Forest, and XGBoost were trained on both an open-source Kaggle dataset and the custom dataset generated through our IoT system. Among these, XGBoost consistently demonstrated the highest prediction accuracy, lowest regression error, and strongest correlation with actual RUL values. Random Forest showed competitive performance, while SVR and KNN exhibited lower accuracy due to their limited ability to capture nonlinear degradation behavior.

The results validate that combining real-time IoT sensing with data-driven machine learning provides a reliable and scalable solution for battery health prognostics. The proposed approach is computationally efficient, hardware-friendly, and capable of being extended to multi-cell battery management systems. Future work can focus on implementing on-board inference using TinyML, incorporating additional sensing modalities such as internal resistance, and adapting the model for different battery chemistries and operating profiles.

VI. Future Work

There are many areas of improvement for this project. First, there is potential to expand the system to have additional capabilities to recognize a greater range of dynamic gestures and two-handed signs, as well as apply continuous sequence recognition, possibly utilizing temporal models like LSTM or 3D CNNs. In addition, datasets could be improved in diversity to capture a larger number of users and environments, providing a better model for real-world signals and improving generalizability. Second, if we can leverage NLP such that syntactic trees are created from recognized signs or paths, then meaningful sentences can be formed - which would support two-way communication. Lastly, it would be interesting to optimize the model for deployment on mobile or edge devices using tools like TensorFlow Lite or ONNX in order to expand accessibility and potentially contribute to real-world settings where resources are constrained.

References

[1] P. Sharma and B. J. Bora, "A Review of Modern Machine Learning Techniques in the Prediction of Remaining Useful Life of Lithium-Ion Batteries," *Batteries*, vol. 9, no. 2, pp. 1–25, 2023.

[2] A. Thelen et al., "Uncertainty-Aware and Explainable Machine Learning for Early Prediction of Battery Degradation Trajectory," *Digital Discovery*, Nature Publishing Group, 2024.

[3] Y. Han et al., "Denoising Transformer-Based Neural Network for RUL Prediction," *Energies*, vol. 16, no. 4, 2023.

[4] C. Ji et al., "A Review on Lithium-Ion Battery Modelling from Mechanism to Machine Learning," *Processes*, vol. 12, no. 1, 2024.

[5] Z. Jin et al., "A Data-Driven Framework for Lithium-Ion Battery RUL Using Hybrid Models," *Applied Energy*, Elsevier, 2024.